Artificial Intelligence & Machine Learning – Task 3

Model Validation, Overfitting Control & Hyperparameter Tuning

Project: Enhanced House Price Prediction System

Dataset: California Housing Dataset

Author: Shivangi Arya

1. Objective and Methodology

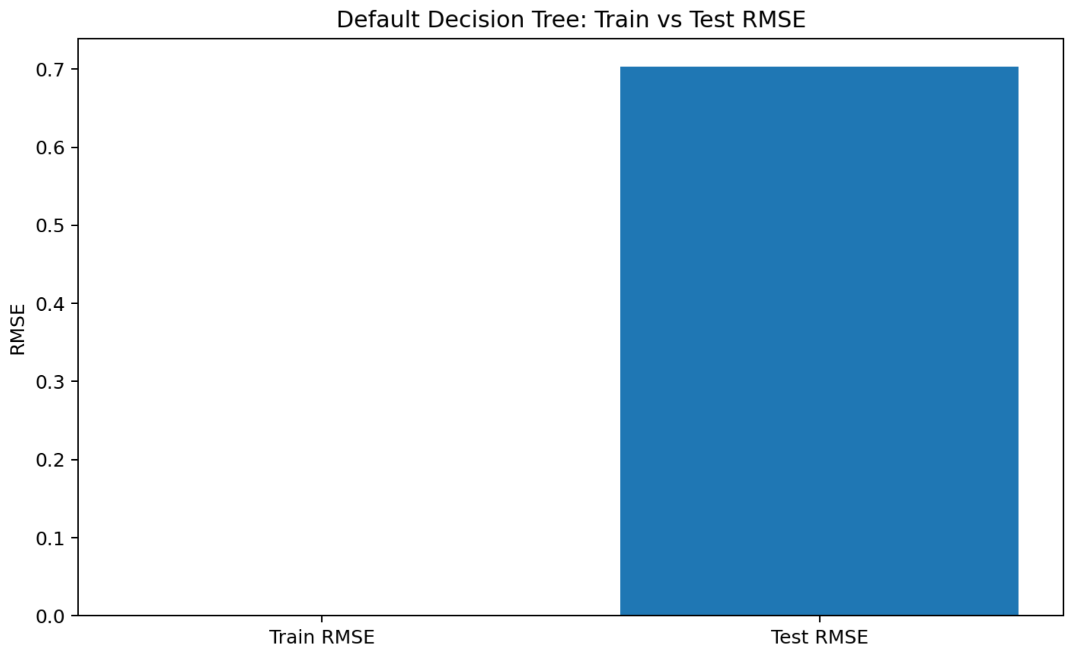
Task 3 extends the Task 2 house-price prediction system by focusing on model validation, overfitting control, systematic hyperparameter tuning and generalization. The same California Housing dataset and 80/20 train-test split with random_state=42 were used. StandardScaler is placed inside scikit-learn pipelines so preprocessing is fitted only on training data and within cross-validation folds.

The analysis first evaluates an unrestricted Decision Tree to expose overfitting. Five-fold cross-validation is then applied to the training data. Finally, GridSearchCV evaluates max_depth values of 3, 5, 7 and 10 together with min_samples_split values of 2, 5 and 10, using negative RMSE as the scoring metric.

2. Model Performance

Model	RMSE	R^2
Linear Regression	0.7456	0.5758
Ridge Regression	0.7456	0.5758
Tuned Decision Tree	0.6454	0.6822

The unrestricted Decision Tree had training RMSE 0.0000 and test RMSE 0.7037. Its training R^2 was 1.0000, compared with a test R^2 of 0.6221. The near-perfect training fit and substantially weaker test performance demonstrate overfitting.

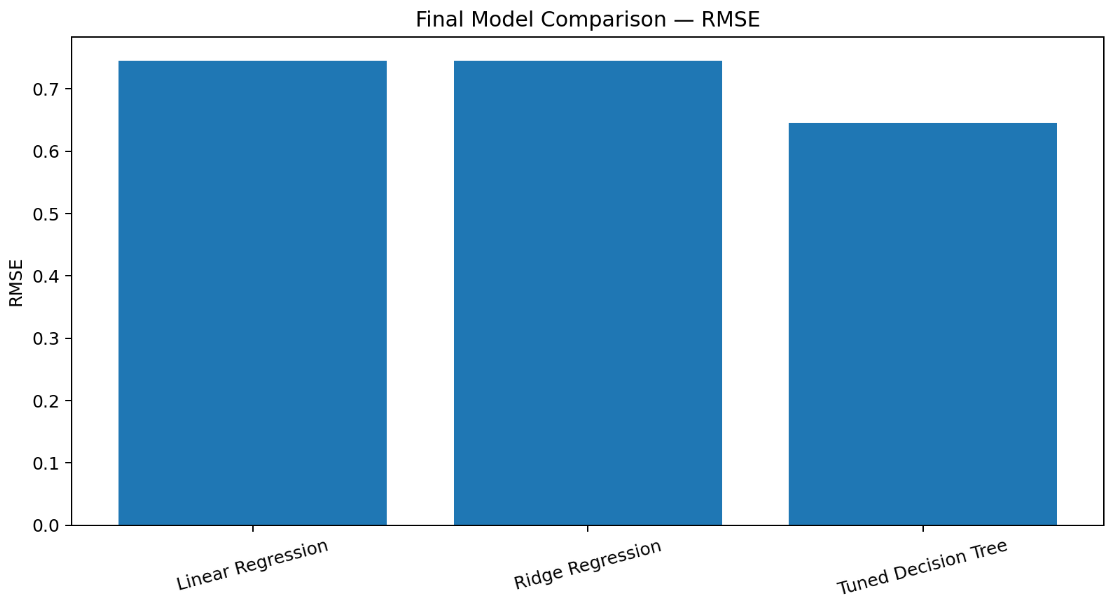


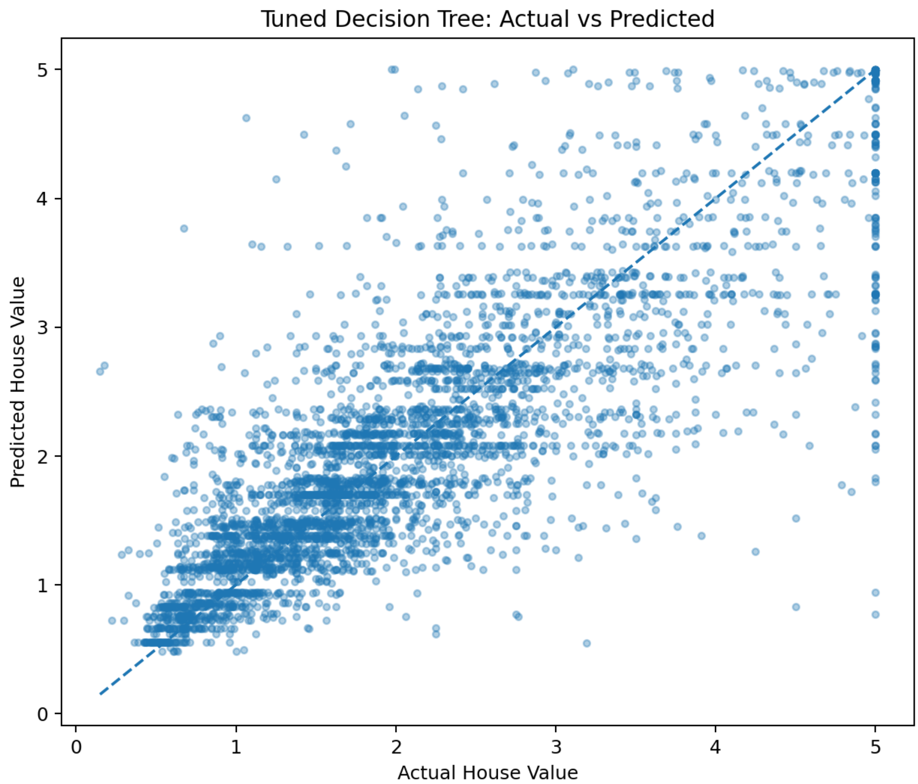
3. Cross-Validation and Hyperparameter Tuning

Five-fold cross-validation of the unrestricted tree produced a mean RMSE of 0.7240 with a standard deviation of 0.0199. GridSearchCV selected max_depth=10 and min_samples_split=10. The best cross-validation RMSE was 0.6375.

The tuned model achieved a test RMSE of 0.6454 and test R^2 of 0.6822. Its training RMSE was 0.4804, giving a test-vs-train RMSE gap of 0.1650. This is a much healthier fit than the unrestricted tree and indicates that the depth/split constraints successfully reduce memorization of the training data.

4. Final Model Validation





The tuned Decision Tree is the strongest model among the tested alternatives because it has the lowest held-out RMSE and the highest R^2. Compared with the Task 2 Decision Tree result (RMSE 0.7242, R^2 0.5997), the tuned model improves held-out RMSE to 0.6454 and R^2 to 0.6822.

5. Conclusion and Model Selection Justification

The final model is the tuned Decision Tree Regressor. The selection is supported by three pieces of evidence: (1) the unrestricted tree showed clear overfitting; (2) five-fold cross-validation provided a more reliable generalization estimate than a single training fit; and (3) GridSearchCV systematically tested the specified hyperparameter combinations and found max_depth=10 and min_samples_split=10 to be the best configuration by RMSE. The optimized model also outperformed the Task 2 Linear Regression and Ridge baselines on the same test set.

This workflow satisfies the Task 3 goal of moving beyond a single train/test score toward reproducible validation, controlled model complexity and evidence-based hyperparameter selection.